
CSL4010: ML-DLOps

Assignment 1

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Submission:

- **Google Colab:**
<https://colab.research.google.com/drive/1lgo45Z4CEL3kgqr8-b8Ee8q448CEa8Ba?usp=sharing>
- **GitHub Repository:**
<https://github.com/diwanshuydv/MLOps-Diwanshu-Yadav-B23CS1017/tree/Assignment-1>

1 Comparative Evaluation of ResNet Architectures (Q1-a)

1.1 Experimental Configuration

This study evaluates the performance of ResNet-18 and ResNet-50 on the MNIST and FashionMNIST datasets. The datasets were partitioned into training, validation, and testing sets with a **70/10/20** split ratio. To maintain parity, both models were trained from scratch (`pretrained=False`) with variations in batch sizes (16, 32), optimizers (Stochastic Gradient Descent and Adam), and learning rates.

1.2 Results: FashionMNIST Performance

FashionMNIST presents a higher degree of intra-class variance compared to standard digits. The empirical results are documented in Table 1.

Table 1: Classification Accuracy on FashionMNIST Dataset

Batch Size	Optimizer	LR	ResNet-18 Acc (%)	ResNet-50 Acc (%)
16	SGD	10^{-3}	91.43	89.84
16	SGD	10^{-4}	85.28	78.17
16	Adam	10^{-3}	91.61	91.09
16	Adam	10^{-4}	92.26	91.79
32	SGD	10^{-3}	90.17	88.23
32	Adam	10^{-4}	89.61	90.09

1.3 Results: MNIST Performance

MNIST represents a relatively low-complexity feature space. Accuracy metrics are detailed in Table 2.

Table 2: Classification Accuracy on MNIST Dataset

Batch Size	Optimizer	LR	ResNet-18 Acc (%)	ResNet-50 Acc (%)
16	SGD	10^{-3}	99.01	98.89
16	SGD	10^{-4}	97.15	96.09
16	Adam	10^{-3}	98.70	97.20
16	Adam	10^{-4}	99.29	99.26
32	SGD	10^{-3}	98.96	98.42
32	Adam	10^{-4}	98.80	98.90

1.4 Analysis and Discussion

1. **Optimization Synergy:** The Adam optimizer consistently converged to a higher accuracy optima compared to SGD across both datasets. Notably, Adam displayed superior robustness at lower learning rates (10^{-4}), where SGD performance significantly degraded.
2. **Architectural Depth vs. Feature Complexity:** ResNet-18 frequently matched or exceeded the performance of ResNet-50. This suggests that for 28×28 grayscale images, the additional depth of ResNet-50 provides diminishing returns and may introduce optimization challenges (e.g., vanishing gradients or overfitting) due to the simplicity of the input manifold.
3. **Learning Rate Sensitivity:** A learning rate of 10^{-4} coupled with Adam provided the highest generalization performance, suggesting that smaller weight updates are beneficial for these specific classification tasks.

2 Kernelized Support Vector Machine (SVM) Analysis (Q1-b)

The performance of non-linear SVMs was evaluated using Polynomial (**poly**) and Radial Basis Function (**rbf**) kernels.

Table 3: SVM Kernel Performance and Computational Latency

Dataset	Kernel	Test Accuracy (%)	Training Time (ms)
MNIST	Poly	97.71	181,106
MNIST	RBF	97.92	167,134
FashionMNIST	Poly	86.30	283,958
FashionMNIST	RBF	88.28	232,496

2.1 Analysis

- **Kernel Superiority:** The RBF kernel outperformed the Polynomial kernel in both accuracy and training efficiency. This indicates that the Gaussian kernel provides a more effective mapping for the underlying data distributions.
- **Dataset Complexity:** A distinct accuracy gap (approx. 10%) is observed between MNIST and FashionMNIST, reinforcing the finding that fashion items require more sophisticated feature extraction than handwritten digits.

3 Hardware Acceleration and Computational Profiling (Q2)

This section quantifies the acceleration provided by GPU computing and the theoretical computational cost measured in Floating Point Operations (FLOPs).

Table 4: Hardware Performance Profiling (Batch Size: 16, FashionMNIST)

Compute Device	Model	Test Acc (%)	Train Time (ms)	FLOPs
GPU (CUDA)	ResNet-18	86.38	73,544	66,368,000
	ResNet-50	82.75	153,237	157,510,144
CPU	ResNet-18	85.52	656,216	66,368,000
	ResNet-50	81.86	1,478,222	157,510,144

3.1 Analysis of Hardware Performance

- **Throughput and Acceleration:** The transition from CPU to GPU provided a substantial performance gain. For ResNet-18, the GPU achieved a speedup of approximately **8.92x**. For the more complex ResNet-50, the GPU speedup was approximately **9.65x**, demonstrating that hardware acceleration becomes increasingly critical as model depth and FLOPs increase.
- **Computational Complexity:** As evidenced by the FLOPs count, ResNet-50 is roughly **2.37x** more computationally expensive than ResNet-18. This correlates directly with the observed training times on both hardware backends.
- **Observation on Accuracy:** Interestingly, ResNet-18 outperformed ResNet-50 in accuracy on this specific task. This is likely due to the limited resolution of the FashionMNIST input (28×28), where the deeper architecture of ResNet-50 may suffer from overfitting or optimization difficulties compared to the leaner ResNet-18.